Zhiyuan Li

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Education

- **University of Toronto** — Toronto, Ontario
  BASc in Engineering Science, Robotics; Cumulative GPA: 3.90/4.00 — *Sep 2022 – Apr 2027*

Research Experience

- **University of Toronto Robot Vision and Learning Lab** — Toronto, Ontario
  Undergraduate Researcher, supervised by Prof. Florian Shkurti — *May 2026 – Present*
  - Designed an **episodic memory module for frozen VLA policies** ($\pi_{0.5}$) that stores **context, plan, failure, and recovery** in one entry to support plan selection and failure recovery on memory-dependent manipulation tasks (**RMBench**); analyzed which failure modes cross-episode memory can and cannot fix.
  - Undergraduate thesis on **agentic learning from failed attempts** in code-as-policy robotics, where a frozen **coding agent** converts rollout videos, API traces, and scores into verified, **failure-specific progress checks**. Evaluating on **RoboTwin 2.0** and **EmbodiedSWE-Bench**, with a **22-percentage-point improvement** in task success on RoboTwin 2.0 so far.
- **Huawei Technologies, Noah's Ark Lab — Embodied AI** — Toronto, Ontario
  Research Assistant — *May 2025 – Dec 2026*
  - Major contributor to The North Star's **BEHAVIOR Challenge** system: **1st place at CoRL 2026** (Q score 0.683; 48.2% full-task success on 100 household tasks), after **4th place at NeurIPS 2025**. Built the **agentic harness**, whose capability- and history-based **policy cards** route tasks among VLA policies and a **skill library** of FSM pipelines combining SAM6D, $\pi_{0.5}$, and TAMP. Built the **HiPhys-RSI** proof of concept, where a frozen-VLM **coding agent** composes tools into code pipelines and revises them from rollout videos and logs, and validated it for **whole-body control** and **long-horizon manipulation**.
  - Conducted large-scale robustness evaluations of **VLAs** ($\pi_{0.5}$, X-VLA) and **world action models** (LingBot-VA, Fast-WAM, GE-Act, Cosmos-Policy) on **LIBERO-Plus**, and built **RoboTwin 2.0-Plus**, a dual-arm benchmark with the same perturbation protocol. Analyzed how **vision** (camera viewpoint, lighting, background, sensor noise), **language** (paraphrased instructions), **object layout** (distractors), and **robot initial-state** perturbations affect each model family: WAMs are more robust to noise, lighting, and layout (74.2% vs. 58.6% for $\pi_{0.5}$ on RoboTwin 2.0-Plus), while camera and initial-state shifts remain hard. Deployed each model with matched action chunking (16–50 steps), action spaces, and denoising settings.
  - Led the design of two **benchmarks** for open problems in VLA and WAM evaluation, **safe manipulation under clutter** and **anticipation of future dynamics**: **RoboPRO**, 80 bimanual tasks with 16K demonstrations and **safety-aware contact metrics**; and **RoboDyna**, 30 **dynamic** tasks with **synchronized and asynchronous** evaluation protocols that isolate the effect of **inference latency**. Found that clutter reduces success and increases non-target collisions, and that latency mainly changes which rollouts succeed rather than overall success, with systematic losses on time-critical tasks. Accepted at the **NeurIPS 2026** and **RSS 2026** workshops.
  - Proposed the **Agentic Adversarial Harness ($\mathbf{A^2H}$)**: a coding agent designs task variations that break its own robot programs, repairs them, and distills the failures into a **reusable harness that transfers to unseen environments**, raising real-robot **first-attempt success** from 15% to 60% over a conventional self-improvement harness. Co-developed **RoboHarness**, which orchestrates VLA, WAM, RL, and TAMP policies in sequence as agentic skills with **memory-guided handoffs**, reaching 77.2% success on **RMBench** and 43% on six **BEHAVIOR-1K** tasks. A^2H was accepted at **CoRL 2026** (Late Breaking Results); RoboHarness received the **Best Paper Award** (oral) at the ECCV 2026 Workshop on Agent in World.
  - For the **Self-CriTeach** project, fine-tuned LLMs for **long-horizon symbolic planning** on self-generated **PDDL** domains: converted solver plans into **chain-of-thought self-teaching** traces for SFT, then applied self-critiquing constrained RL with goal-predicate rewards. Compared training recipes: CoT vs. symbolic-only traces, SFT vs. RL vs. SFT+RL, and distillation, majority-voting, and self-distillation baselines. Raised **Qwen3-4B** planning success from 26% to 46% (above GPT-4o at 19%); on a real **UR5e** robot it reached 70% vs. 40% for a PDDL solver, showing greater robustness to imperfect perception. Accepted at **ICML 2026**.
- **University of Toronto Continuum Robotics Laboratory** — Toronto, Ontario
  Undergraduate Research Assistant (Part-time), supervised by Prof. Jessica Burgner-Kahrs — *Feb 2024 – Apr 2025*
  - Studied state estimation for **tendon-driven continuum robots (TDCR)**, extending the **Clarke transformation** to general TDCR models to characterize actuation constraints and degrees of freedom.
  - Derived **TDCR forward kinematics** from first principles and built **MuJoCo** simulations; compared them with physical robots and introduced **corrective factors** for hardware effects such as material stiffness, disk radius, and tendon length. Documented results for the **OpenCR** project.

Publications

Conference Papers

- *Learning to Anticipate Physical Failures via Agentic Adversarial Harness.*
  Z. Li, D. Liu, A. Rasouli.
  CoRL 2026, Late Breaking Results.
- *Self-CriTeach: LLM Self-Teaching and Self-Critiquing for Improving Robotic Planning via Automated Domain Generation.*
  J. Huang, **Z. Li**, Y. Hu, Z. Zhang, M. Coates, X. Quan, Y. Zhang.
  ICML 2026.
- *Distracted Robot: How Visual Clutter Undermine Robotic Manipulation.*
  A. Rasouli, M. Alban, S. Pakdamansavoji, **Z. Li**, Z. Zhang, A. Wu, X. Zhao.
  ICRA 2026.

- *Improving Robotic Manipulation Robustness via NICE Scene Surgery.*
  S. Pakdamansavoji, M. Pourkeshavarz, A. Sigal, **Z. Li**, R. H. Yang, A. Rasouli.
  ICRA 2026. Earlier version at CoRL 2025 Robot Data Workshop.

Workshop Papers

- *RoboHarness: Memory-Driven Orchestration of Heterogeneous Robot Policies for Long-Horizon Planning.*
  J. Huang, Y. Hu, **Z. Li**, R. Qi, Y. Xiao, Z. Zhang, M. Coates, T. Cao, Y. Zhang.
  ECCV 2026 Workshop on Agent in World, oral. **Best Paper Award**.
- *Benchmarking Robotic Manipulation under Physically Realistic Dynamics.*
  Z. Li, R. H. Yang, X. Zhao, A. Rasouli.
  NeurIPS 2026 Robotics World Modeling Workshop.
- *Data and Evaluation Protocols for Robust Robotic Manipulation.*
  Z. Li, C. Eret, X. Zhao, Y. Wu, A. Rasouli.
  RSS 2026 RoboData Workshop.
- *H-WM: Robotic Task and Motion Planning Guided by Hierarchical World Model.*
  W. Chen, J. Huang, O. Pang, **Z. Li**, X. Hu, L. Zhang, Z. Zhang, M. Coates, T. Cao, X. Quan, Y. Zhang.
  ICLR 2026 Workshop on World Models.

Preprints and Manuscripts

- *Spatial Grafting: Grounding 3D Features for Flow-Matching Robot Policies.*
  D. Liu, Y. Wu, M. Asadi, **Z. Li**, J. Huang, Y. Xiao, T. Cao, Y. Zhang.
  arXiv:2609.35249, 2026.
- *StageGuard: Learning Stage Transitions for Long-Horizon Robot Tasks via Agentic Distillation.*
  J. Huang, Y. Hu, **Z. Li**, R. Qi, Y. Xiao, Y. Wu, T. Ba, Z. Zhang, Y. Zhang.
  arXiv:2609.20791, 2026.
- *How VLAs (Really) Work In Open-World Environments.*
  A. Rasouli, Y. Wu, **Z. Li**, R. H. Yang, X. Zhao, C. Eret, S. Pakdamansavoji.
  arXiv:2604.21192, 2026.
- *Do World Action Models Generalize Better than VLAs? A Robustness Study.*
  Z. Zhang, **Z. Li**, B. Rahmati, R. H. Yang, Y. Ma, A. Rasouli, S. Pakdamansavoji, Y. Wu, L. Zhang, T. Cao, F. Wen, X. Wang, X. Quan, Y. Zhang.
  arXiv:2603.22078, 2026.
- *Mem2Ego: Empowering Vision-Language Models with Global-to-Ego Memory for Long-Horizon Embodied Navigation.*
  L. Zhang, Y. Liu, Z. Zhang, M. Aghaei, Y. Xiao, Y. Hu, M. A. Alomrani, D. G. Arcos Bravo, H. Gu, **Z. Li**, Y. Wu, Z. Zhang, R. Karimi, A. Hamidizadeh, G. Huang, H. Xu, T. Cao, W. Qiu, X. Quan, J. Hao, Y. Zhuang, Y. Zhang.
  arXiv:2502.14254, 2025.

Industry Experience

- **Prelude CA** — Toronto, Ontario
  Full-Stack Engineer / Frontend Lead (Part-time) — *Jun 2025 – Jun 2026*
  - Built **agentic workflows** for a multi-tenant CRM platform: **LLM agents** automatically search the CRM and use **customs declaration data** to find new sales leads. A multi-agent ingestion pipeline matches uploaded CSV columns semantically with **OpenAI embeddings** (90%+ auto-mapping accuracy) and writes validated records to **PostgreSQL**.
- **Yuhang International Flight (Dalian) Co., Ltd.** — Dalian, China
  Full-Stack Engineer — *Jul 2022 – May 2024*
  - Designed and built an **information-management system** for IOU payment records end to end, covering the **SQL database schema**, Python application logic, and a Qt/QML interface. Supported **100+ concurrent users** with connection pooling, bcrypt-hashed authentication, and SQL triggers; indexed fuzzy search returned results in 3–5 seconds over 100k+ records.

Awards

- **1st place, BEHAVIOR Challenge, CoRL 2026** — Team The North Star. 2026
- **Best Paper Award, ECCV 2026 Workshop on Agent in World** — RoboHarness. 2026
- **4th place, BEHAVIOR Challenge, NeurIPS 2025** — Team The North Star. 2025
- **5th place (team), William Lowell Putnam Competition** — top 3 at University of Toronto; individual rank 154 (top 4%). 2023
- **1st place, University of Toronto Undergraduate Mathematics Competition.** 2023
- **Participant, Canadian Mathematical Olympiad.** 2019, 2021
- **Honour Roll, CEMC mathematics contests** — Euclid, Fermat, Hypatia, and CSMC. 2021 – 2022

Skills

- **Model Training**: PyTorch, Hugging Face; VLA fine-tuning ($\pi_{0.5}$), LLM fine-tuning, diffusion and flow-matching policies; multi-GPU training on Slurm clusters
- **Robotics**: UR5e, Franka, and AgileX robots with ROS/ROS2; TAMP and PDDL planning; simulators (Isaac Sim, MuJoCo, SAPIEN, PyBullet); Nav2, V-SLAM, 6D pose estimation (SAM6D), OpenCV
- **Agents**: Claude Code, Codex, OpenAI and Anthropic APIs, LangChain/LangGraph; code-as-policy pipelines and agentic self-improvement loops
- **Software**: Python, C/C++, MATLAB; React, Next.js, Node.js, PostgreSQL, Qt; Linux, Git, Docker

Academic Service

- **Reviewer**: IEEE/RSJ IROS 2026; IEEE ICRA 2027.